

Pandas Cheat Sheet: Real-Life Functions

Category	Function	Key Parameters	What It Does
Load Data	pd.read_csv(filepath, sep=';', usecols=None, nrows=None)	filepath, sep, usecols, nrows, dtype	Load CSV file
	pd.read_excel(filepath, sheet_name=0)	sheet_name, usecols, dtype	Load Excel sheet
	pd.read_sql(query, con)	query, con	Load from database
	pd.DataFrame(data)	data (dict, list, array)	Create DataFrame manually
Explore	df.head(n=5)	n	First rows
	df.tail(n=5)	n	Last rows
	df.shape	-	Rows × Columns
	df.info()	-	Column types + nulls
	df.describe()	include, percentiles	Summary stats
Select/Filter	df['col']	col name	Select column
	df[['c1','c2']]	list of cols	Select multiple columns
	df.loc[rows, cols]	row labels, col labels	Select by label
	df.iloc[rows, cols]	row index, col index	Select by position
	df[df['col'] > x]	condition	Filter rows
Clean Data	df.dropna(axis=0, how='any')	axis, how, subset	Drop missing values
	df.fillna(value)	value, method	Fill missing values
	df.drop(columns=['c1','c2'])	columns	Drop columns
	df.rename(columns={'old':'new'})	columns	Rename columns
	df.astype({'col': type})	dtype mapping	Change datatype
	df.duplicated()	subset, keep	Check duplicates
	df.drop_duplicates()	subset, keep	Remove duplicates

	df.replace(old, new)	to_replace, value	Replace values
Aggregate & Group	df.groupby('col')	by, as_index	Group rows
	df['col'].sum()/mean()/count()	-	Aggregations
	df.pivot_table(values, index, columns, aggfunc)	values, index, columns, aggfunc	Pivot table summary
	df['col'].value_counts()	normalize, dropna	Count unique values
Merge/Join	pd.concat([df1, df2], axis=0)	axis, ignore_index	Stack vertically/horizontally
	pd.merge(df1, df2, on='col', how='inner')	on, left_on, right_on, how	SQL-style join
	df.join(other_df, how='left')	how	Join on index
Apply Functions	df.apply(func, axis=0)	func, axis	Apply across rows/cols
	df['col'].map(func)	function/dict	Apply to one column
	df['col'].apply(lambda x: ...)	custom function	Custom transformation
Sort & Rank	df.sort_values('col', ascending=True)	by, ascending	Sort rows
	df.sort_index()	ascending	Sort by index
	df.rank()	method, ascending	Rank values
Time Series	pd.to_datetime(df['col'])	format, errors	Convert to datetime
	df.set_index('date')	inplace	Set datetime index
	df.resample('M').mean()	rule (D,M,Y), aggfunc	Resample
	df['date'].dt.year/month/day	-	Extract parts
Export Data	df.to_csv('file.csv', index=False)	index, sep, encoding	Save to CSV
	df.to_excel('file.xlsx', sheet_name='Sheet1')	sheet_name, index	Save to Excel
	df.to_sql('table', con, if_exists='replace')	name, con, if_exists	Save to DB